

Learning by Quiz Creation in Programming Education: Proposal and Experimental Application of Generative Artificial Intelligence (AI)-based Approach

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Abstract: With the rapid advancement of information technology these days, there is an increasing demand to cultivate human resources capable of identifying and solving issues effectively through information technology, further underscoring the importance of information education. In this context, “learning by quiz creation,” which learners create quizzes on their own, is expected to be highly effective in fostering the application of knowledge and skills. While there have been many studies on methods to support learning by quiz creation, this study proposes a new approach to support learning by quiz creation using generative artificial intelligence (GenAI). We designed quiz creation processes (QCPs) for three typical types of quiz creation (story-, quiz-, and solution-based) aimed at deepening learners’ understanding. Furthermore, we developed quiz creation prompts to enable GenAI to guide these QCPs effectively while also supporting learners’ proactive quiz creation. To verify the effectiveness of the proposed method, we set two research questions: (1) Can GenAI adequately guide the proposed QCPs? and (2) Can GenAI with prompts enhance learning effectiveness of quiz creation? We conducted an experimental application involving university students, and the results suggest that GenAI can serve as a suitable guide for learners in the initial stages of each QCP, with particular effectiveness observed in solution-based quiz creation. Additionally, GenAI equipped with quiz creation prompts received higher evaluations from learners compared to general GenAI, demonstrating its potential to enhance the learning experience.

Keywords: quiz creation, generative artificial intelligence, programming education

1. Introduction

1.1 Growing Importance of Information Education

With the continual growth in the use of information technology, the development of human resources with appropriate information utilization skills is important in Japan to prepare for the arrival of Society 5.0 [1]. The Ministry of Education, Culture, Sports, Science and Technology (MEXT) has organized the qualities and abilities constituting information utilization ability, including the ability to effectively discover and address issues by using information technology appropriately. In particular, the Computers and Programming unit of Information I—a course on information in Japanese high schools—requires students to develop and implement purpose-specific algorithms, utilize computers and networks through programming, and evaluate and improve processes. However, cultivating these abilities in actual educational settings is difficult because the knowledge retained by novice programmers is often superficial, and there are high barriers to applying it to actual problem solving. Robins et al. [2] organize the problems faced by students new to programming, including problems in the transition from theory to practice and stages of modeling the issue and its representation. Robins et

al. observed that students have already learned but cannot use fragile knowledge; to develop the abovementioned abilities, understanding and using this knowledge in a deeper way is essential. While research has been conducted on various aspects of programming, such as the use of gamification [3], [4], flipped learning [5], and project-based learning [6], the specific methods of education that should be used remain the subject of debate.

1.2 Learning by Quiz Creation

Learning by quiz creation, in which learners create their own quizzes, necessitates a deeper understanding of the quizzes and solutions than quiz-solving; it is expected to be more effective for learning [7], [8]. We believe that through quiz creation, learners can develop the ability to apply their knowledge and skills to finding and solving problems, and this is highly compatible with the abilities required in contemporary information education.

Since the inception of learning by quiz creation, numerous related studies have been conducted, including a survey and analysis of quiz creation activity [9], [10] and an analysis of quizzes generated by learners [8]. Various learning support systems for quiz creation have been developed, including those aimed at evaluating the quizzes created by other learners [11], [12], collaborative learning using the quizzes created [13], and automatic diagnosis of the created quizzes by simplifying the quiz creation process (hereinafter “QCP”) through the integration of single sentences [14]. QSIA (Questions Sharing and Interactive

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Assignments)[11], QPPA (Question-Posing and Peer Assessment learning system)[12], CollabTest[13], enable flexible quiz creation, including the “natural language expression of quiz setting”[8]. However, because no function directly supports the QCP itself, learning outcomes tend to vary with each learner’s level of understanding or their own QCP.

Meanwhile, Monsakun [14] simplifies the QCP, allowing for process control and automatic diagnosis of the quiz. Specifically, the system employs an approach in which quizzes are built by combining pre-prepared single-sentence cards; this design is intended to create an environment where quiz creation learning—generally considered cognitively demanding—can be practiced even in elementary education, particularly in lower-grade mathematics classes. However, this simplification of the QCP like Monsakun limits the range of knowledge that can be handled. Therefore, the issue is that the learning units and quiz formats that can be created are limited.

Our survey found no existing work that completely overcomes these issues. In this study, while addressing the challenges outlined above, we set separate learning objectives for each of the three typical types of quiz creation defined by Hirashima et al.[14], targeting the cultivation of learners’ ability to discover and solve problems through the use of information technology. Our aim is to provide learners with support that will enable them to achieve these objectives.

The details of three typical quiz creation types will be discussed later.

1.3 Generative AI

Recently, interactive generative AI (GenAI; hereinafter “GenAI”), such as ChatGPT, has gained attention for its potential use in education. GenAI uses large language models trained by deep learning to generate responses; it offers natural dialog capabilities, high versatility, and multimodal support. GenAI is expected to improve the efficiency of various educational and learning activities, lower learning costs, and facilitate self-learning. Especially, in the field of programming learning, the use of GenAI is said to be effective, because it shows high performance in various programming tasks, including source code completion and correction, automatic syntax error correction, and text-to-code generation [15].

Examples of expected educational applications of GenAI include math tutors, automated essay grading, and dialog partners for learning English [16]; several studies have been conducted on GenAI applications in programming education. Jonsson et al. [17] conducted a pedagogical application of natural language programming among university students using the GPT-3 Codex, proposing that the discrepancies and errors generated by GenAI could be utilized as a source of friction to encourage learners to contemplate novel concepts and to engage in creative trial and error. Yilmaz et al.[18] investigated the impact of utilizing GenAI tools on student’s computational thinking abilities, programming self-efficacy, and learning motivation. Their findings indicated that GenAI was particularly efficacious for enhancing computational thinking and self-efficacy. Liu et al.[19] investigated the efficacy of CS50.ai, which implements

GenAI to furnish students with tutoring-like learning assistance in a CS50 course at Harvard University; their findings indicated that the AI tool enhanced student comprehension through coding explanation and style improvements. Nevertheless, no examples of the direct application of GenAI to the learning process of quiz creation were identified in our research.

1.4 Purpose and Research Question

This study aims to foster learners’ ability to discover and address issues using information technology through quiz creation. Thus, the proposed method supports quiz creation concerning programming learning utilizing GenAI. First, we present QCPs, which have been designed to enhance the learning effectiveness of quiz creation for programming learning. These are based on the three typical types of quiz creation [14]. Second, we present quiz creation prompts enabling GenAI to smoothly guide learners along the proposed QCPs. These prompts facilitate supporting learners’ QCPs, enhance the precision of GenAI’s output using CoT, and encourage learners to engage in proactive learning. We used GPT-4, an OpenAI-developed text generation model.

The primary contribution of this study is that it supports abilities that are currently considered important in information education through learning by quiz creation. The originality of this study is three QCPs and the quiz creation prompts to adapt GenAI to learning by quiz creation. QCPs ensure effective quiz creation that enhances learners’ understanding, and the prompts allow GenAI to guide the QCPs and be an effective assistant to the students. Learning by quiz creation, in which students create quizzes and their answers, rather than simply solving the given problems, contributes to the essential development of information utilization abilities.

Based on the originality of this study, we set two research questions in this paper:

- (1) Can GenAI adequately guide the proposed QCPs?
- (2) Can GenAI with quiz creation prompts enhance learning effectiveness of quiz creation?

Herein, we examined the effectiveness of learning by quiz creation using GenAI, based on the student’s interaction history with GenAI in quiz creation, the quizzes created through the interaction, pre- and post-tests conducted before and after the experiment, and the questionnaire results. This paper extends the research presented at TALE 2024 [20].

2. Quiz Creation Process

We examined the QCPs that can achieve high learning effectiveness by quiz creation. We consider three QCPs based on Hirashima’s [14] three typical quiz creation types (story-, quiz-, and solution-based).

Following the examples of QSIA [11], QPPA [12], and CollabTest [13] discussed in Section 1.2, we foresee two primary applications for the created quizzes: (i) fostering collaborative learning by having students peer-evaluate one another’s quizzes and (ii) using the more polished quizzes as comprehension test for learners. As a preliminary step, this paper proposes method aimed at deepening learners’ understanding by supporting the



Prepared by the author based on the definition of the quiz by Hirashima et al[14].

Fig. 1 Definition of the quiz by Hirashima et al.

Table 1 Quiz creation types and learning objectives.

Type	Learning objective
Story	Be able to use programming knowledge to discover real-world issues
Quiz	Be able to explain the relationship between programming knowledge and its advantages and disadvantages
Solution	Be able to apply specific knowledge in programming to solving real-world problems

QCP itself.

2.1 Three Typical Quiz Creation Types

Hirashima et al. [14] define a general quiz in learning as shown in **Fig. 1**. Given information (GI) is the information that can be treated as known when taking a quiz, whereas required information (RI) is the information that can be deduced from GI. Meanwhile, the procedure for deriving RI from GI is called solution method (SM). In this way, a quiz is defined by GI, RI, and SM, and various quiz creation tasks can be defined by imposing constraints on them. There are three types of quiz creation tasks, depending on the constraints on each of the quiz's three components, as well as three types based on these tasks: story-, quiz-, and solution-based.

Story-based quiz creation is creating quizzes using GI as a constraint, such as a picture or a story, and then creating quizzes based on it. Meanwhile, quiz-based quiz creation uses GI and RI as constraints. The original quiz is administered, and students create new quizzes by partially modifying it. Lastly, solution-based quiz creation is creating a quiz with SM constraint. Learners create quizzes that allow for the use of specific solution methods.

Table 1 shows the learning objectives for the three types of quiz creation for the high school information course “Information I” assumed in this study. To establish the objectives, we refer to the description in the “Courses of Study for Senior High Schools” in Japan [1] and the characteristics of each quiz creation type described earlier.

2.2 Proposal of the Quiz Creation Processes

To achieve the learning objectives shown in Table 1, we examined the QCPs for each of the three quiz creation types, as illustrated in **Fig. 2**. Each of the three QCPs is carried out through interaction with GenAI. Our aim is to overcome the limitation identified in learning support systems such as QSIA and CollabTest, as discussed in Section 1.2: they lack functionality for supporting the QCP itself. By advancing quiz creation interactively with GenAI while remaining within the guardrails provided by the QCP, we seek to ensure that every learner, regardless of their level of understanding, can obtain the substantial learning benefits of quiz creation.

As indicated in **Fig. 2**, the first step of each quiz creation process—“present the story,” “present the quiz,” and “present the solution”—begins with the teacher giving learners a quiz creation task that includes specific constraints (story, quiz, and solution). **Table 2** illustrates, as examples, the quiz creation tasks for each quiz creation type that were presented in the experiment conducted for this study. When setting quiz creation tasks such as those shown in Table 2, teachers need to carefully design each task while paying attention to the information that needs to be constrained within the given quiz creation type—GI, RI, or SM. The details of what quiz creation tasks should be assigned in each quiz creation type are described in Sections 2.2.1 to 2.2.3 below.

In the subsequent steps, the QCP progresses by repeatedly following a routine in which GenAI poses questions that encourage learners to carry out the step proactively, and the learners respond. To make the flow of the QCP more concrete, we present a dialogue between a student and GenAI taken from the experimental application in Section 5.2. This example not only demonstrates the basic routine described above but also, in part, shows how each QCP step explained below is actually implemented.

All three QCPs include the “Diagnose/Correct” of the learner-created quiz and the “Summary of dialog.” In the “Diagnose/Correct” step, GenAI diagnoses the learner's quiz and encourages the learner to correct it if necessary. The diagnostic standards include “logical relevance of the quiz content,” “appropriateness of the answer to the quiz,” “sufficiency of GI in the quiz content,” “readability and operability of source code,” and “typographical errors and omissions.” This step has two objectives. First, to improve the completeness of the quiz. As noted at the beginning of this chapter, we plan to eventually leveraging the quizzes created by learners as comprehension test. In this case, it is essential that the created quizzes be free of ambiguous wording and errors. A quiz that has been created by learners and then diagnosed and revised according to the criteria above will, as a result, allow other learners to engage with the problem more smoothly. Furthermore, when learners peer-evaluate these quizzes, the ensuing discussion is expected to focus on the quiz content itself rather than on superficial issues such as typographical errors. Second, the process deepens learners' understanding. We consider the iterative act of refining a quiz to make it better to improve the learners' understanding of the knowledge and skills covered in the quizzes.

In the “Summary of dialog” step, the learner and GenAI summarise their dialogue and reflect on the learning. Reflection on learning is essential for deepening learners' understanding. In our proposed method, learners not only carry out reflection on their own but also receive feedback on that reflection from GenAI. Because GenAI generates its output while retaining the prior dialogue, its feedback can offer learners fresh insights into their own learning. By reflecting together with GenAI in this step, we believe the overall learning effect of quiz creation can be further enhanced. We will now explain the three QCPs, focusing on each step in relation to the learning objectives.

2.2.1 Story-based

In story-based quiz creation, the teacher first assigns a quiz

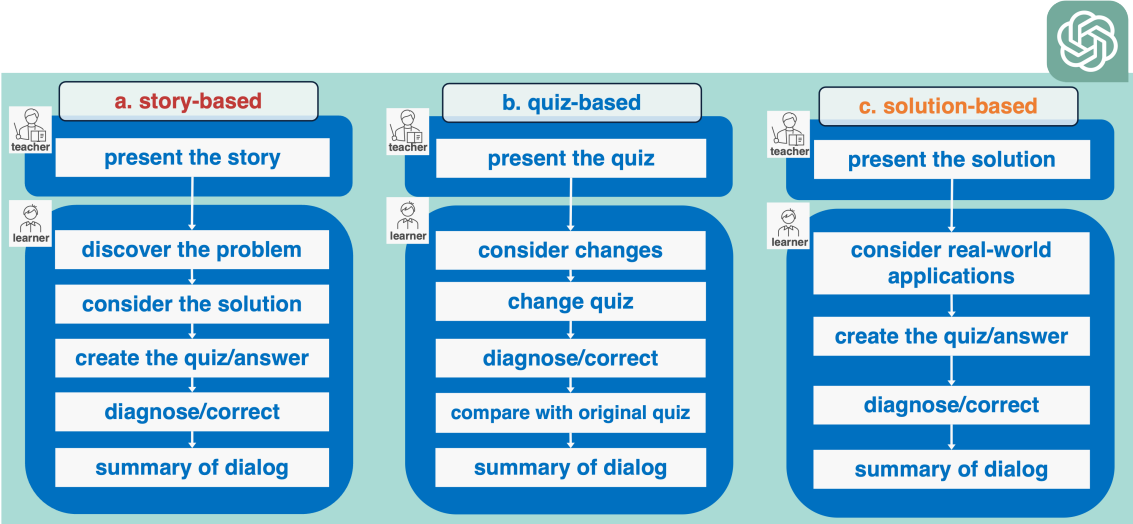


Fig. 2 Proposed quiz creation process.

Table 2 Example of quiz creation task (used in the experiment described in Section 4).

Type	Quiz creation task
Story	Create a programming-related quiz based on the information system that supports Emi’s activities described below.
	(Story) In the library where Emi works, managing the lending status of books is an issue. A significant operational challenge for libraries is that a considerable number of patrons do not return books after the due date. Checking for overdue items and notifying the patrons concerned are presently carried out manually, which further increases staff workload. Emi wonders whether this repetitive task can be automated.
Quiz	Change part of the following quiz and answer to create a new quiz.
	(Original Quiz) Here is the following list: data = [59, 44, 43, 49, 42, 75, 100, 38, 91, 3, … , 11, 86, 80, 90, 5, 67, 23, 60, 97, 29]; Write a source code to sort this list in ascending order using the quicksort algorithm. (Answer) def swap(a, i, j): # Function to swap two elements in the list. a[i], a[j] = a[j], a[i] ...
Solution	Create a quiz to deepen the understanding of the concept of “class” in object-oriented programming by dealing with everyday life subjects that can be solved by making classes.

creation task that includes a story. In a story-based quiz creation task, the story must contain a problem that can be solved by applying programming knowledge or concepts. In the example of a story-based quiz creation task shown in Table 2, the character’s problem is designed to be solvable chiefly through the concept of “control structures”. However, teachers should avoid explicitly inserting programming terms and concepts—for instance, “loops” or “conditional branches” in this case—into the story itself. This restraint gives learners the opportunity to determine for themselves where a programming-solvable problem exists in the story and how, in concrete terms, the relevant knowledge and concepts can be applied to resolve it. Consequently, learners are expected to acquire the perspective needed to discover problems that can be solved with programming knowledge. The learner extracts the RI (problem) from the story and then examines the SM for that RI (problem). When the learner thoroughly examines the SM, he/she creates a quiz using the RI (problem) and an answer based on the SM. GenAI diagnoses the created quizzes and answers, and the learner

corrects them as needed. After making the corrections, the learner reviews the dialog during the QCP.

The proposed story-based QCP allows learners to simulate the process of discovering and solving a problem based on a given story, utilizing their programming knowledge. This may enable learners to enhance their abilities to discover real-world issues from the aspect of information technology.

2.2.2 Quiz-based

Quiz-based quiz creation begins with the teacher presenting the quiz creation task, which includes the original quiz and the answer to the source code. In a quiz-based quiz creation task, the quiz and its solution should ideally deal with knowledge or concepts that have many variants within the same category—for example, sorting algorithms such as “merge sort” and “quick sort”, as illustrated in Table 2. This is because in the quiz-based quiz creation, learners create a new quiz by partially modifying the given quiz and answer. By incorporating various types of knowledge as described above, a wider variety of options can be considered when modifying quizzes. As a

result, more active discussions are expected in collaborative learning settings, which we plan to implement in the future. Furthermore, through quiz-based quiz creation utilizing such knowledge and concepts, learners can not only individually deepen their understanding of related knowledge, but also be encouraged to develop a more systematic understanding that includes the interrelationships among them. This includes, for example, understanding differences in characteristics such as advantages and disadvantages compared to other concepts, as well as differences in applicable domains.

The answer is generally not presented in quiz-based quiz creation [14], but the proposed method does so. This is due to the concept of source code optimization in programming, and we expect that source code optimization can also be a programming learning experience. The learner considers which sections of the quiz and the answer can be changed. Then, he/she changes the quiz/answer based on the considerations, and GenAI determines whether those changes are appropriate. Based on the diagnosis, learners correct the changed quiz as needed. Subsequently, the learner compares the changed quiz to the original. The comparison should focus on how other learners' learning on each quiz differs. Finally, the learner reflects on the dialog that occurred during the QCP.

These processes may allow learners to develop the ability to associate and understand the variety of programming knowledge and choose the most appropriate method for a given situation.

2.2.3 Solution-based

Solution-based quiz creation begins with the teacher presenting the quiz creation task, including the solution's constraints. The solution must incorporate knowledge related to programming. In addition, the proposed method recommends restricting the use of "everyday life topics." The aim is that, by creating quizzes rooted in familiar contexts, learners will develop the ability to apply their programming knowledge to real-world problem solving. On this premise, instructors should choose concepts that are readily transferable to daily situations—such as "classes" in object-oriented programming, as illustrated in Table 2—and that match learners' levels of understanding. By contrast, concepts like pointers/addresses or recursion may feel remote from the day-to-day experience of novice programmers (e.g., the high-school students considered in this study). Accordingly, such concepts should not be selected as the solution in these circumstances. The learner considers the real-world applications of the presented solutions. When the real-world application of the solution has been sufficiently considered, the learner creates a quiz and an answer based on the considerations. GenAI then diagnoses the quiz/answers, and the learner corrects them as needed. Finally, the learner reviews the dialog during the QCP.

Solution-based quiz creation allows learners to better understand a specific SM by creating quizzes that incorporate that method. They can also develop a high level of application ability by simultaneously considering "how to apply this method to daily life."

3. Quiz Creation Prompts

Next, we designed quiz creation prompts for each of three quiz

Table 3 Prompt format for quiz creation.

Role
You are a professional programming instructor.
Order
Please assist students in learning programming.
Constraints
· The programming language should be Python.
Input
(Include specific input content here)

Table 4 Prompt for story-based quiz creation (partial excerpt).

Constraints
⋮
· If it is necessary to present the source code in the dialog, please limit the initial explanation to a simple flow using pseudocode, etc.
· Even if the creator requests specific code, do not provide all of it; leave important parts of the algorithm blank to encourage the creator to think.
· If you think it is necessary during the dialog, feel free to ask the creator many questions.
· Conduct the dialog according to the following flow:
1. Discovery of programming-related problems included in the survey presented story:
2. Consideration of solutions to the selected task:
3. Creation of the quiz and answer:
4. Diagnosis and correction of the created quiz and answer:
5. Summary of the dialog:
⋮

creation types. In designing the prompts, we aimed to use GenAI to guide learners through the QCPs proposed in Section 2. We designed the quiz creation prompt to enable teachers to modify only specific parts of the prompt, thereby allowing learners to create questions for a wide range of knowledge domains merely by adjusting a few specific sections. This flexibility aims to overcome the constraint—identified in Section 1.2 for learning support systems such as Monsakun—that quiz creation is restricted to a narrow range of subjects and knowledge areas. **Table 3** shows the format of the quiz creation prompts. OpenAI, which developed ChatGPT, has introduced several methods to improve ChatGPT outputs [21]. Among them, this prompt format is designed based on three points: "Include details in your query to get more relevant answers," "Ask the model to adopt a persona," and "Use delimiters to clearly indicate distinct parts of the input." Furthermore, we show the main points we considered when designing the prompts for creating quizzes.

- (1) Assume that the learners are new to programming and explain politely and clearly.
- (2) Ensure that the source code generates only parts of the whole, such as pseudocode or fill-in-the-blank formats, giving learners opportunities to think.
- (3) Divide the designed QCPs into parts and explain each one to ensure that the dialog follows the process.
- (4) Make the learners the main actors in each step of the process, with GenAI serving as an assistant, providing helpful questions or hints.
- (5) Utilize CoT (Chain-of-Thought) [22] to enable step-by-step explanations and suggestions.

Table 4 shows a portion of the designed quiz creation prompts. Table 4 is an excerpt from the prompt "# Constraints" for the story-based quiz creation, and it primarily shows prompts related to points (2), (3), and (4). More detailed explanations follow

the excerpt for each step of the QCP. The “# Input” section describes the main part of the quiz creation task: the story for story-based, the original quiz for quiz-based, and the solution for solution-based. When the prompt is put into practice, teachers configure the quiz creation task to suit the specific knowledge domain and document the task details in this section. As a result, it becomes possible to provide learning environment of quiz creation that accommodates a broad spectrum of topics and subjects.

4. Experimental Application to University Students

An experiment was carried out to assess the effectiveness of the proposed QCPs and quiz creation prompts. The subjects were 12 fourth-year undergraduate students at the University of Electro-Communications. They were majors in information science and technology and, through the courses they had completed, had acquired basic knowledge of information technology, including programming.

Table 5 shows the flow of the day of the experiment. The pre-test was administered before the day of the experiment. Its content was designed separately for each quiz creation types based on the learning objectives shown in Table 1 and the quiz creation tasks shown in Table 2. **Table 6** outlines the contents of the tests. All students entered the experiment after completing the three pre-tests. To equalize the mean scores of those pre-tests across the cohort, students were divided into three groups of four; the resulting allocation was announced under “Grouping” in Table 5.

The quiz was created three times, in the order of story-based, quiz-based, and solution-based. During the “Description of quiz creation task” step for each quiz creation activity in Table 5, the quiz creation task shown in Table 2—one tailored to each quiz

creation type—was presented, and the learners proceeded to carry out the task accordingly.

Each group created quizzes in the following three patterns through the three types of quiz creation, and **Table 7** shows the correspondence between the quiz creation patterns and each group. The same quiz creation task was presented shown in Table 2 to every student in every type, regardless of quiz creation pattern (NN, YN, YY).

- NN: Without using GenAI (only using internet research)
- YN: Using GenAI (without quiz creation prompt setting)
- YY: Using GenAI (with quiz creation prompt setting)

In pattern YY, the students pasted the quiz creation prompts as their initial prompt and then proceeded with the quiz creation, freely interacting with GenAI. We used BetterChatGPT [23] as the environment for interacting with GenAI. BetterChatGPT is an open-source chatbot (third-party) that allows users to import/export dialog history and prompt libraries, and to set ChatGPT parameters. **Figure 3** shows an example of the BetterChatGPT screen.

A post-test and a questionnaire were administered after the quiz creation. The content of the post-test was the same as the pre-test, and the questionnaire was mainly conducted to investigate the achievement of learning objectives and what students think about learning by quiz creation itself or using GenAI to create quizzes.

5. Results

5.1 Implementation Status of the Proposed QCPs

To evaluate whether GenAI with the quiz creation prompts was able to properly guide the proposed QCPs, we examined the dialog histories of pattern YY students in each of the three

Table 7 Correspondence between students and patterns in each quiz creation.

Student ID	A, B, C, D	E, F, G, H	I, J, K, L
1st (Story-based)	YY	NN	YN
2nd (Quiz-based)	YN	YY	NN
3rd (Solution-based)	NN	YN	YY

Table 5 Experiment flow.

	Time (minute)	Contents
Preparation	5	Obtain consent
	5	Research description
	10	Preferences for GenAI
	5	Grouping
Quiz creation 1 (Story-based)	5	Description of quiz creation task
	30	Quiz creation
	5	Collect quiz, answer, dialog history
	30	Post-test and questionnaire
Quiz creation 2 (Quiz-based)	5	Description of quiz creation task
	30	Quiz creation
	5	Collect quiz, answer, dialog history
	30	Post-test and questionnaire
Quiz creation 3 (Solution-based)	5	Description of quiz creation task
	30	Quiz creation
	5	Collect quiz, answer, dialog history
	30	Post-test and questionnaire

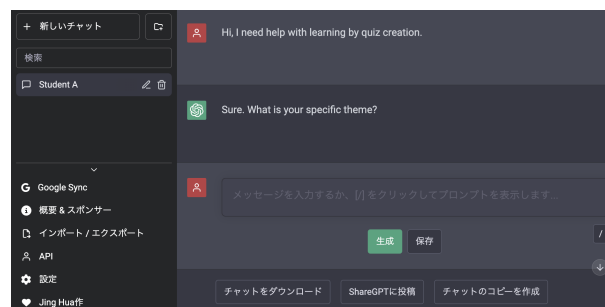


Fig. 3 Example of BetterChatGPT screen.

Table 6 Summary of questions in the pre- and post-test.

Type (PA total)	No (PA)	Content of the question
Story (25)	1–3 (5×3)	Write programming on control structures
	4 (10)	Find a problem in the presented story that can be solved by programming and create a programming quiz based on it
Quiz (30)	1 (10)	Take multiple sorting algorithms and compare algorithms based on their characteristics
	2 (10)	Consider elements that can be changed for a given quiz about sorting algorithm
	3 (10)	Select one of the elements listed in Q2, change it, and explain how learners' learning of solving it will change
Solution (20)	1 (10)	Solve a given real-world issue by applying classes in object-oriented programming
	2 (10)	Describe the characteristics of object-oriented programming (as compared to procedural programming)

quiz creation types (4 per type, 12 in total) to determine whether the designed QCPs was implemented. The two criteria for “implemented” are as follows

- (1) that there is question from GenAI to ensure smooth implementation of the process; and
- (2) that the learner is proactively implementing the process in response to GenAI’s guidance.

Table 8 through **Table 10** summarize the evaluation results. These tables suggest that the process was generally well-guided in the early stages. Meanwhile, as the process progressed, the number of instances where guidance was effectively maintained decreased.

5.2 Dialog History and Created Quizzes

Table 11 and **Table A-1** show a portion of the dialog history of a student in the experiment and the quiz generated through that dialog. Student D in **Table 11** and **Table A-1** conducted the story-based quiz creation with pattern YY.

Based on **Table 11**, we describe how each step of the story-based QCP shown in Fig.2 was carried out. As noted in Section 2.2, our approach assumes a routine in which GenAI poses questions that drive the process, and learners advance the QCP by responding. No.2, 4, 6, and 7 in **Table 11** are such messages from GenAI, and immediately after each prompt student D replies, thereby moving the process forward. No.2 and 3 illustrate the “Discover the problem” step of the story-based QCP: encouraged by GenAI to identify a problem faced by the story’s characters, he identifies the core problem as the fact that “the loan history is not structured as data.”

Next, No.4 and No.5 shown in **Table 11** indicates that the “Consider the solutions” step was being executed. GenAI poses questions that encourage student D to consider how to solve the problem in the story that he identified during the “Discover the Problem” step—specifically which data the lending history should contain and how those data should be combined to form an

Table 8 Implementation of each process for students who created quiz in pattern YY (story-based).

Student ID	A	B	C	D	%
Discover the problem	×	○	○	○	75
Consider the solutions	○	○	○	○	100
Create the quiz/answer	×	○	○	○	75
Diagnose/Correct	○	×	×	○	50
Summary of dialog	○	×	×	×	25

○ : Implemented, × : Not implemented

Table 9 Implementation of each process for students who created quiz in pattern YY (quiz-based).

Student ID	E	F	G	H	%
Consider changes	○	○	○	○	100
Change quiz	×	○	○	○	75
Diagnose/Correct	×	×	○	○	50
Compare with original quiz	○	○	○	×	75
Summary of dialog	×	×	×	×	0

○ : Implemented, × : Not implemented

Table 10 Implementation of each process for students who created quiz in pattern YY (solution-based).

Student ID	I	J	K	L	%
Consider real-world applications	○	×	○	○	75
Create the quiz/answer	○	×	○	○	75
Diagnose/Correct	○	×	×	○	50
Summary of dialog	×	×	×	×	0

○ : Implemented, × : Not implemented

Table 11 Dialog history of student D (story-based, YY).

No.	Speaker	Content
1	Student D (Pasted prompt)	# Role You are a professional programming instructor... # Input ... The confirmation of overdue books and notifications to users are currently done manually, which increases the workload. Emi is considering whether this repetitive task can be improved through some form of automation.
2	AI	Let us think about Emi’s issue at the library. What problems could be solved using programming?
3	Student D	Regarding the library’s issue, I believe we first need to structure the lending history into a data structure. I think this data structuring issue should be presented as a sub-quizzes.
4	AI	That is a great idea! Structuring the lending history is the first step in solving the problem. Specifically, what data should we include, and how should we combine these data to create the data structure?
5	Student D	I think it would be good to create a class for the lending history that has the following attributes: ·Lender ·Lending time ·Due date (year/month/day/hour)
6	AI	That is wonderful! A class representing the lending history needs attributes such as the lender, lending time, and due date. Based on this information, let us create a quiz and an answer. What kind of quiz will you create? ...
7	AI	That is an excellent answer! By giving the class attributes such as borrower’s name, e-mail address, lending time, lending period, and return deadline, you have properly represented the lending history. However, what data types do you assume for <code>lend_time</code> and <code>due</code> ? In addition, will the calculation <code>lend_time + due</code> operate correctly for those data types? Clarifying these points will make the quiz statement even easier to understand.
8	Student D	In Python, how should I handle time formats such as year/month/day/time?
9	AI	In Python, we use the <code>datetime</code> module to handle time. This module provides the <code>date</code> class for dates, the <code>time</code> class for times, and the <code>datetime</code> class for both date and time. ...
10	AI	That’s a great answer! By creating an instance of the <code>Lending</code> class, you can enter specific loan history. However, with this answer alone it is still unclear how the lending history is managed and how notifications are sent. To clarify these points, how about setting the next sub-question as “Create a class to manage lending history and implement a method inside it to send notifications”?
11	Student D	I’ll adopt your suggestion: Quiz 3 will be “Create a class that manages lending history and implement within it a method for sending notifications.”

appropriate data structure. Student D responded to this question by proposing to define a class that holds data related to the lending history, with attributes for the person who borrowed the item, the date of borrowing, and the borrowing deadline. Subsequently, in No.6, GenAI prompted student D to create an actual quiz and answer based on the dialogue so far.

From No.7 to No.11 shown in Table 11 indicates that the “Diagnose/Correct” step was being executed. After student D entered question and answer which created on his own into GenAI, in No.7, GenAI pointed out an issue in a specific attribute of the answer code. When he asked about the comment in No.8, GenAI immediately replied, as shown in No.9. In No.10 and No.11 GenAI proposed an additional sub-question, which he decided to adopt.

In the example featuring Student D, we did not observe implementation of the “Summary of dialog” step. In that step, however, we anticipate that GenAI will pose prompts such as “What new insights did you gain through this quiz creation?” The learner would then reflect on the quiz creation experience, and GenAI would subsequently provide feedback on that reflection.

Through the dialogue based on the given story, student D eventually created a quiz in which the character’s problem is solved using classes in object-oriented programming. Because of time constraints in this experiment, student D could not prepare an answer to Quiz 3. Nevertheless, that question was intended to “implement a method within the class that manages the lending history”, which was expected to require control structures. **Table 12** shows an overview of the quizzes created for each quiz creation type and the quiz creation patterns of the students who created them.

5.3 Pre- and Post-test

Table 13 summarises the results of the three pre- and post-tests introduced in Table 6. For each quiz creation pattern, it lists the mean score and standard deviation, together with the numbers of students whose post-test scores increased, remained unchanged, or decreased relative to their pre-test scores. To examine whether the score differences between creation patterns (NN, YN, YY) vary within the same quiz creation type, we applied the Mann–Whitney U test for each format (story-, quiz-, and solution-based) and for each pattern combination (NN-YN, NN-YY, YN-YY) at a significance level of 5%. Taking multiple comparisons into account, we set the significance level for each test at $\alpha = 0.05/3 \approx 0.017$ using the Bonferroni method. However, no statistically significant differences were found in any of the combinations. We also examined differences in the proportions of students whose scores improved using Fisher’s exact test with the same combinations at the 5% significance level. It should be noted that the labels “No change” and “Down” in Table 13 were treated as a single label, “No up,” in this test. Again, no statistically significant differences were observed in any combination.

5.4 Responses to the Questionnaire

Figure 4 shows the results of the questionnaire that asked whether the learners felt that using GenAI for learning by

Table 12 Summary of quizzes created for each quiz creation type.

Type	Summary of created quizzes
Story	- Pseudo-implement a function that determines which users are nearing or past their due dates and sends periodic notifications using control structures (NN, YN, YY) - Implement a function to create a new list from the book loan list that contains only the books that are overdue and the users who have borrowed them, by means of a control structure (NN, YN, YY) - Implement a database to automate library tasks using classes (YN, YY) - Implement a function that uses a control structure to determine and flag books that are overdue based on data frames that store lending information (YY)
Quiz	- Implement multiple algorithms for sorting a list and observe how the time complexity varies by algorithm (NN) - Implement a sorting algorithm executed by a specified time complexity (NN) - Change the sorting method of the list from ascending to descending order (NN) - Consider the algorithm itself, including the change in the number of elements in the unsorted portion and the change in time complexity due to pivot selection (NN, YN) - Implement quick sort using a method other than the given solution (YN) - Implement sorting algorithms other than quick sort (YN, YY) - Change the order of the original list and select/implement the fastest algorithm for the features of the list’s order (YY) - Implement a sorting algorithm based on the specified pivot selection method (YY)
Solution	- Implementing game character parameters using classes (NN) - Reproduce videotape operations with class (NN) - Implement a menu management system for a cafe using classes (YN) - Implement a management system for school student information using the class (YN) - Implement a class that manages a baseball team and implement a class that manages information on each player as its inner class (YY) - Implement classes containing methods for loading data, calculating statistics, etc. to analyze test results (YY)

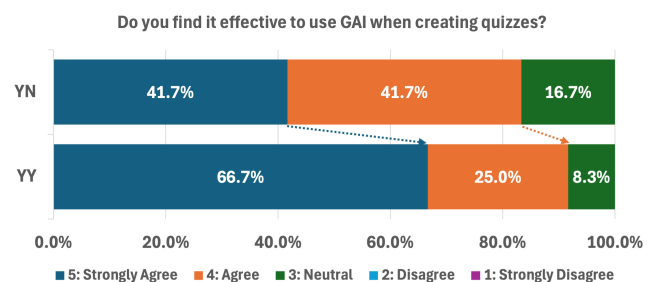
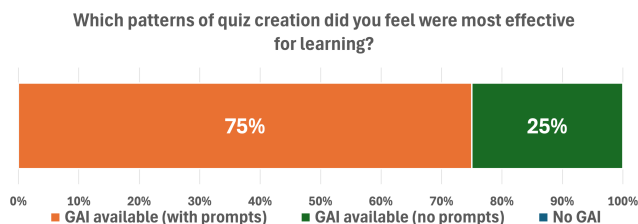


Fig. 4 Questionnaire results (1).

quiz creation was effective, and **Fig. 5** shows the results of the questionnaire that asked which quiz pattern (NN, YN, YY) they felt was the most effective after all quiz creation. In addition, **Table 14** and **Table 15** shows some of the reasons for the answers to the questions in Fig. 5. Figure 4 illustrates that 83.4% of students in Pattern YN and 91.7% in Pattern YY expressed a positive opinion when asked “Do you find it effective to use GenAI when creating quizzes?”. Additionally, 41.7% of students in Pattern YN and 66.7% in Pattern YY selected the “5: Strongly

Table 13 Pre- and post-test results.

Type (PA total)	Pattern	Pre-Test		Post-Test		Score Change		
		Mean	SD	Mean	SD	Up	No-up	
							No-change	Down
Story (25)	NN	16.50	2.69	17.00	2.92	1	3	0
	YN	13.00	4.64	12.00	7.97	2	0	2
	YY	14.75	6.94	16.75	7.22	2	2	0
Quiz (30)	NN	13.25	8.98	14.50	5.50	2	0	2
	YN	14.50	9.18	15.75	5.76	2	1	1
	YY	12.25	7.40	18.00	1.87	3	1	0
Solution (20)	NN	8.25	6.65	8.25	5.31	1	2	1
	YN	11.25	3.77	13.25	2.49	3	0	1
	YY	9.50	6.06	12.50	5.50	4	0	0

**Fig. 5** Questionnaire results (2).**Table 14** Reasons for selecting pattern YY.

GenAI with prompt allowed users to focus on the necessary parts of quiz creation because the rail was laid for quiz creation. It was also designed to make users think, so I could think for myself without relying too much on GenAI.

GenAI instantly told me what I needed to work on and what I needed to ask during the activity. The conversation-based programming made it smooth for me to create the source code.

I found GenAI with prompt to be highly effective for learning, as it helped me to consolidate my programming skills and to work step-by-step to accomplish tasks.

First, I liked being able to interact with AI. It was nice to be able to pinpoint knowledge I did not understand and not have to spend a lot of time generating code and data sets. GenAI with prompt would have been better; it suggested “what to ask” so the learning went smoothly.

GenAI with prompt explained each step in a way that even I, a beginner, could understand and gave me hints to help me understand.

Learning while interacting with GenAI is more advantageous in that questions can be answered immediately and specific examples can be provided immediately. I felt that when we freely talk with GenAI, it immediately creates the answer to the quiz, and learners end up on the side of asking questions to understand the answers, but with the pre-prompts, they can be on the side of answering questions, which allows them to think more spontaneously. I also thought it was good that the next step is clearly stated in an easy-to-understand manner, allowing the learner to proceed with the questions as GenAI tells them, and giving them a sense of active learning.

In the case of GenAI with prompt, GenAI already understood the task, and I liked that it praised my proposal but pointed out areas for improvement.

In the case of GenAI without prompt, I did not feel like I learned because GenAI came up with all the code and output on its own; I appreciated that GenAI with prompt railed me through the problem step by step. I also liked that it gave me a chance to think for myself.

Agree” option in response to the question. Furthermore, Fig. 5 illustrates that when asked “Which patterns of quiz creation did you feel was most effective for learning?”, 25% of the students responded that it was “GenAI available (without prompts)”, while 75% indicated that it was “GenAI available (with prompts).” Overall, GenAI with the quiz creation prompts resulted in higher post-learning satisfaction for the learners compared to GenAI without the quiz creation prompts.

Table 15 Reasons for selecting pattern YN.

I felt that I could get more information that I wanted to know if I set my own GenAI requirements.

GenAI with prompt was completed more quickly, but I could not create the quiz freely because the prompt set the direction of quiz creation. This was fine for teaching fixed content, but I thought it might eliminate flexible thinking on the part of those being taught.

From a complete beginner’s perspective, I felt that GenAI without prompt was more efficient. In the case of GenAI without prompt, the answers to the quiz are provided at the beginning, and I felt that I was able to understand the results of changing the quiz based on those answers through trial and error. In the case of GenAI with prompts, I felt that the learning progressed without knowing the goal, and therefore, the understanding was not very deep. However, I felt that GenAI with prompt is best suited for those who have a certain level of understanding to further their learning.

6. Discussion

6.1 RQ1: Can GenAI Adequately Guide the Proposed QCPs?

Analysis of the dialog histories suggests that the quiz creation prompts guided learners smoothly through the early phases of the QCP. However, the implementation rate of the later steps tended to decrease. All of the cases that were judged “not implemented” in the later steps shared a single cause: they failed to satisfy Criteria (1)—“that there is guidance question from GenAI to ensure smooth implementation of the process”—as defined in Section 5.1. In this study, the role of the quiz creation prompt is to guide learners smoothly along the QCP by posing questions that allow them to advance the process proactively. The above result suggests that, as the dialogue deepens, the constraints that the initial quiz creation prompt imposed on GenAI diminish, and consequently GenAI stops asking the questions needed to execute the process—i.e., the effect of the prompt weakens. To address this, we intend to reset the dialog screen and reload the prompts at each process stage in subsequent attempts.

The results also confirm the flexibility of GenAI in learning by quiz creation. In the story-based quiz creation of this experiment we had assumed that the learner would create a quiz concerning “control structures”. However, as Table A-1 shows, Student D finally created the quiz in a context where the character’s problem is solved by applying the concept of classes in object-oriented programming. On the other hand, the sub-quiz (Quiz 3) required implementing a method using control structures, and its content was proposed by GenAI. These results indicate that GenAI possesses high flexibility, dynamically switching its output to match each learner’s focus and utterances while still ensuring that

the main knowledge specified by the task (control structures in this case) is addressed.

6.2 RQ2: Can GenAI with Prompts Enhance Learning Effectiveness of Quiz Creation?

The efficacy of GenAI in learning by quiz creation is evaluated through an assessment of the pre- and post-test scores and the responses to the questionnaire.

First, we examined, for each quiz creation type, whether score differences between the pre- and post-tests differed among quiz creation patterns. The Mann–Whitney U tests revealed no statistically significant differences in any format or pattern. We then performed Fisher’s exact tests on the proportion of students whose scores improved, but again found no significant differences. However, because each creation pattern had only four students in this experiment, these test results are expected to be strongly affected by the small sample size. Therefore, caution is required in interpreting the statistical reliability of these results.

On the other hand, the fact that every student in pattern YY in the solution-based quiz creation achieved higher post-test scores than pre-test scores serves as supplementary evidence of the effectiveness of the proposed method. This results indicates that the learning gains were greater when the solution-based quiz creation was conducted using GenAI with quiz creation prompts than when quiz creation was conducted using internet research alone. In solution-based quiz creation, the SM is typically presented as the sole constraint within the context of the quiz creation task. However, in this experiment, to achieve the learning objective set for solution-based quiz creation, we imposed the additional constraint of “Dealing with everyday life issues” in addition to the constraint of the SM. Because of it, solution-based quiz creation in this experiment might be more challenging than the other two quiz creation types, which may have hindered learners’ ability to achieve optimal learning outcomes independently. The test results suggest that GenAI with quiz creation prompts may act as an effective assistant in more challenging solution-based quiz creation. Consequently, it may have contributed to raising learners’ ability “to find and solve issues utilizing information technology” to a more advanced level.

The results shown in Table 14 indicate that many learners found learning by quiz creation using GenAI with prompt to be most effective because it laid down the “rails” for executing the QCP, allowing learners to focus on the necessary parts and making quiz creation more effective, and the questions and step-by-step hints provided by GenAI were found to give learners more opportunities to think for themselves and promote a proactive attitude toward learning without relying too much on AI. These considerations suggest that GenAI with the quiz creation prompts more effectively supports learning by quiz creation and improves learners’ comprehension. In addition, the results shown in Table 15 indicate that students who favored GenAI without prompts gave the following reasons: they felt that GenAI without prompts gave them more freedom and made it easier to obtain the information they wanted to know, and it first provided them with examples of quiz creation so that they could

deepen their understanding through trial and error of adjusting them. The results of these analyses will be used as material to further improve GenAI prompts.

7. Conclusion

The objective of this study was to foster learners’ ability to find and solve issues utilizing information technology through learning by quiz creation. To achieve this, we proposed a QCP to achieve high learning effectiveness and quiz creation prompts to appropriately guide the QCPs using GenAI. To evaluate the effectiveness of these methods, we conducted an experimental application of learning by quiz creation using the proposed methods on 12 university students.

Analysis of the student’s dialog history suggested that the proposed quiz creation prompts effectively guide implementation of the QCP’s initial stages. We examined, for each quiz creation type, whether score differences between the pre- and post-tests differed among quiz creation patterns. We applied the Mann–Whitney U test and Fisher’s exact test to the pre- and post-test data, but neither test revealed statistically significant differences. However, these results are likely to have been strongly affected by the small sample size, so their statistical reliability should be interpreted with caution. Conversely, in the solution-based condition, every student who carried out quiz creation under Pattern YY scored higher on the post-test than on the pre-test. We regard this fact as a useful supplementary evidence of the effectiveness of the proposed method. Based on this and the fact that the solution-based quiz creation in this experiment was more challenging than the other two types, it is suggested that GenAI with quiz creation prompts can effectively act as an assistant to learners in tackling advanced tasks and take their ability “to find and solve issues utilizing information technology” to a more advanced level. The questionnaire after the experiment revealed that learners felt more effective when using GenAI with quiz creation prompts than GenAI without it, suggesting that the use of GenAI with quiz creation prompts can improve learners’ understanding more.

On the other hand, this study has limitations. The experiment dealt with different domain knowledge in each quiz creation format (control structures, sorting algorithms, classes), so we cannot rule out the possibility that domain-specific backgrounds or proficiency differences influenced the results. Although the participants were selected on the assumption that they possessed comparable understanding of these three domains, future work should compare the quiz creation types under unified domain conditions to quantitatively evaluate domain dependence.

In the future, we will examine methods to guide QCPs more precisely. In addition, through more detailed analysis of the qualitative data obtained in this experiment, such as dialog history, we plan to examine the QCP for further deepening learners’ understanding and the quiz creation prompts to further improve learning effectiveness.

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Appendix

Table A-1 Quiz and answer created by student D.

	Content
Quiz	Main Quiz In the library where Emi works, managing the lending status of books is an issue. In particular, many users do not return books even after the due date, which poses a major challenge for the library's operations. The confirmation of overdue books and notifications to users are currently done manually, which increases the workload. Let's try to solve this problem using programming.
	Sub-quiz 1 To automate the confirmation of overdue books and notifications to users, we first need to structure the lending history into a data structure. Let's use the concept of classes in programming to structure the lending history.
	Sub-quiz 2 Define the lending history class and input the lending history. Sub-quiz 3 Make a class to manage the lending history and implement a method to send notifications within that class.
Answer	<pre> # Quiz 1 from datetime import datetime, timedelta # Lending History Class class Lending: def __init__(self, name, mail, lend_time_y, lend_time_m, lend_time_d, lend_time_h ↪ , lend_time_min, lend_time_sec, due): self.name = name # Name of the person who borrowed self.mail = mail # Email address of the person who borrowed self.lend_time = datetime(lend_time_y, lend_time_m, lend_time_d, lend_time_h, ↪ lend_time_min, lend_time_sec) # Time of borrowing self.due = due # Borrowing period self.due_time = self.lend_time + timedelta(days=self.due) # Due return time # Quiz 2 lending1 = Lending(name="John", mail="john@example.com", lend_time_y=2022, ↪ lend_time_m=1, lend_time_d=1, lend_time_h=13, lend_time_min=30, lend_time_sec ↪ =0, due=7) # Quiz 3 # Not created due to time constraints </pre>



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